Formalising Pédrot's Prefascist CwF with Π-Types in Agda

Hok Ip Chow[1]

Advisor: Liang-Ting Chen[2]

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[1]FLoV, University of Gothenburg, Gothenburg, Sweden

[2]Institute of Information Science, Academia Sinica, Taipei, Taiwan

Syntax vs Semantics

There are two broad approaches to studying a dependent type theory.

- **Syntactic**: study syntactic objects (judgements and inference rules) directly, usually by induction on derivations.
- **Semantic**: study syntactic objects by assigning mathematical meanings to them.

Why semantics?

- Prove syntactic properties, such as normalization and decidability of type checking.
- Show that a principle is not derivable by constructing a countermodel.

A **category with families** (CwF) is a standard notion of semantics for dependent type theory.

Categories

A category $\mathcal{C}$ consists of:

- a collection of objects **Obj**;
- for any $x, y : \mathbf{Obj}$, a collection of morphisms $\mathbf{Hom}_{\mathcal{C}}(x, y)$;
- for any $x : \mathbf{Obj}$, an identity morphism $\mathrm{id}_x : \mathbf{Hom}_{\mathcal{C}}(x, x)$;
- for any $x, y, z : \mathbf{Obj}$, a composition operation

$$_ \circ _ : \mathbf{Hom}_{\mathcal{C}}(y, z) \to \mathbf{Hom}_{\mathcal{C}}(x, y) \to \mathbf{Hom}_{\mathcal{C}}(x, z),$$

such that composition is associative: for any $f : x \to y$, $g : y \to z$, and $h : z \to w$,

$$h \circ (g \circ f) = (h \circ g) \circ f,$$

and identities are indeed identities: for any $f : x \to y$,

$$f \circ \mathrm{id}_x = \mathrm{id}_y \circ f = f.$$

Examples of Categories

- Sets and functions.
- Vector spaces over $\mathbb{R}$ and linear maps.
- The category $\mathbf{1}$ has one object $\star$ and one morphism $\mathrm{id}_\star : \star \to \star$.
- The natural numbers $\mathbb{N}$ with the usual ordering:

$$0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow \cdots$$

  so there is a unique morphism $m \to n$ whenever $m \leq n$.

Contexts and Substitutions

A **context** is a finite list of variable declarations

$$\Gamma = x_1 : A_1, \ldots, x_n : A_n,$$

where later types may depend on earlier variables.

A **substitution** $\sigma : \Delta \to \Gamma$ is a list

$$\sigma = [\, x_1 \mapsto t_1, \ldots, x_n \mapsto t_n \,]$$

such that

$$\begin{aligned} &\Delta \vdash t_1 : A_1, \\ &\Delta \vdash t_2 : A_2[t_1/x_1], \\ &\quad\vdots \\ &\Delta \vdash t_n : A_n[t_1/x_1, \ldots, t_{n-1}/x_{n-1}]. \end{aligned}$$

It replaces every variable of Γ by a term in Δ. Thus, from $\Gamma \vdash A$ type, we obtain $\Delta \vdash A[\sigma]$ type.

Example of a Substitution

Consider the contexts

$$\Gamma = n : \mathbb{N},\ v : \mathsf{Vec}\,\mathsf{Bool}\,n$$

and

$$\Delta = m : \mathbb{N},\ w : \mathsf{Vec}\,\mathsf{Bool}\,(m+1).$$

There is a simultaneous substitution

$$\sigma = [\,n \mapsto m+1,\ v \mapsto w\,] : \Delta \to \Gamma$$

because

- $\Delta \vdash m+1 : \mathbb{N}$;
- after replacing n by $m+1$, $\Delta \vdash w : \mathsf{Vec}\,\mathsf{Bool}\,(m+1)$.

The Category of Contexts and Substitutions

- Every context $\Gamma = x_1 : A_1, \ldots, x_n : A_n$ has an identity substitution

$$\mathrm{id}_\Gamma = [\, x_1 \mapsto x_1, \ldots, x_n \mapsto x_n \,] : \Gamma \to \Gamma.$$

- Substitutions can be composed. Composition is associative and satisfies the identity laws.
- The empty context $\diamond$ declares no variables. For every context Γ, there is a unique empty substitution

$$[\,] : \Gamma \to \diamond.$$

Therefore, contexts and substitutions form a **category with a terminal object**.

Categories with Families

A category with families packages the substitution calculus of dependent type theory.

Starting from the category of contexts and substitutions, it adds:

- for every context Γ, a collection of types $\mathsf{Ty}(\Gamma)$;
- for every $A \in \mathsf{Ty}(\Gamma)$, a collection of terms $\mathsf{Tm}(\Gamma, A)$;
- if $A \in \mathsf{Ty}(\Gamma)$, $t \in \mathsf{Tm}(\Gamma, A)$, and $\sigma : \Delta \to \Gamma$, then

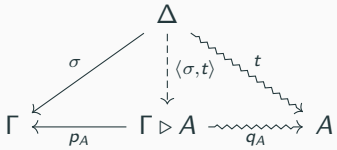
$$A[\sigma] \in \mathsf{Ty}(\Delta), \qquad t[\sigma] \in \mathsf{Tm}(\Delta, A[\sigma]).$$

These operations satisfy the identity and composition laws expected of ordinary substitution.

Context Comprehension

Given a context Γ and a type $A \in \mathsf{Ty}(\Gamma)$, we can form a new context $\Gamma \triangleright A$, which extends Γ with a new variable of type A. It comes with:

- $p_A : \Gamma \triangleright A \to \Gamma$, which forgets the new variable;
- $q_A \in \mathsf{Tm}(\Gamma \triangleright A, A[p_A])$, the new variable;
- for $\sigma : \Delta \to \Gamma$ and $t \in \mathsf{Tm}(\Delta, A[\sigma])$, a unique pairing $\langle \sigma, t \rangle : \Delta \to \Gamma \triangleright A$



$$p_A \circ \langle \sigma, t \rangle = \sigma, \qquad q_A[\langle \sigma, t \rangle] = t,$$
$$\langle p_A, q_A \rangle = \mathrm{id}_{\Gamma \triangleright A}.$$

Presheaves

A presheaf Γ over a category $\mathcal{C}$ consists of the following two assignments:

- for every $x : \mathbf{Obj}$, a set $\Gamma(x)$;
- for every $f : y \to x$, a **restriction** operation $\Gamma(f) : \Gamma(x) \to \Gamma(y)$.

Restrictions preserve identities and composition:

- $\Gamma(\mathrm{id}_x) = \mathrm{id}_{\Gamma(x)}$;
- $\Gamma(f \circ g) = \Gamma(g) \circ \Gamma(f)$.

Notice that the direction of each morphism is reversed.

Question

Let $\mathcal{C}$ be the category $\mathbf{1}$.

What is a presheaf over $\mathcal{C}$?

Trivial

It is simply a set!

Thus, presheaves can be viewed as **variable sets**: a generalisation of ordinary sets.

Question

Let $\mathcal{C}$ be the category $0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow \cdots$.

What is a presheaf over $\mathcal{C}$?

Stages of Computation

Think of n as the number of computation steps we can observe:

$$0 \longrightarrow 1 \longrightarrow 2 \longrightarrow 3 \longrightarrow \cdots$$

A presheaf reverses these arrows:

$$\Gamma(0) \longleftarrow \Gamma(1) \longleftarrow \Gamma(2) \longleftarrow \Gamma(3) \longleftarrow \cdots$$

Here, $\Gamma(n)$ is the set of possible data observable after n steps. A restriction map forgets later information.

Thus, presheaves over this category can be viewed as encoding stages of computation.

Presheaves as CwFs

Presheaves over $\mathcal{C}$ and natural transformations between them form a category $\mathrm{Psh}(\mathcal{C})$.

This category carries a CwF structure:

- Contexts are presheaves and substitutions are natural transformations.
- Types over a context Γ are dependent presheaves over Γ:

$$(x, \gamma) \longmapsto A(x, \gamma), \qquad \gamma \in \Gamma(x).$$

- A term selects an element from each $A(x, \gamma)$, consistently across restrictions.
- Context comprehension is defined pointwise by

$$(\Gamma \triangleright A)(x) = \sum_{\gamma \in \Gamma(x)} A(x, \gamma).$$

Thus, an element of $(\Gamma \triangleright A)(x)$ consists of a context element γ together with a dependent element $a \in A(x, \gamma)$.

Formalising Presheaves

A naïve attempt is to encode the definition directly:

$$
\begin{aligned}
&\textbf{record}\ \mathsf{Psh} : \mathsf{Set}_1\ \textbf{where}\\
&\quad \textbf{field}\\
&\qquad P_0 : \mathsf{Obj} \to \mathsf{Set}\\
&\qquad \mathsf{res} : \forall\{x\,y\} \to \mathsf{Hom}(y, x) \to P_0(x) \to P_0(y)\\
&\qquad \mathsf{res\text{-}id} : \forall\{x\}(\gamma : P_0(x)) \to \mathsf{res}\ \mathsf{id}\ \gamma \equiv \gamma\\
&\qquad \mathsf{res\text{-}comp} : \forall\{x\,y\,z\}(f : y \to x)(g : z \to y)(\gamma : P_0(x)) \to\\
&\qquad\qquad\qquad \mathsf{res}\ (f \circ g)\ \gamma \equiv \mathsf{res}\ g\ (\mathsf{res}\ f\ \gamma)
\end{aligned}
$$

Coherence Hell

The problem is that presheaf equations such as

$$\mathrm{res}\ (f \circ g)\ \gamma \equiv \mathrm{res}\ g\ (\mathrm{res}\ f\ \gamma)$$

are merely propositional.

Hence Agda does not identify the dependent types

$$B(\mathrm{res}\ (f \circ g)\ \gamma) \quad \text{and} \quad B(\mathrm{res}\ g\ (\mathrm{res}\ f\ \gamma))$$

by computation.

Moving terms between them requires explicit transports. These transports accumulate and create further coherence obligations: the **coherence hell**.

Prefascist Sets

Pédrot represents a presheaf by the following "prefascist" record:

record Con : Set_1 **where**

field

$\Gamma_0 : \mathrm{Obj} \to \mathrm{Set}$	raw data
$\Gamma_1 : \forall\{x\} \to \left(\widehat{\gamma} : \forall\{y\}(f : y \to x) \to \Gamma_0(y)\right) \to \mathrm{Prop}$	selects valid thunks

A raw thunk $\widehat{\gamma}$ at x assigns raw data to every morphism into x; Γ_1 is a validity predicate on raw thunks.

Raw-thunk restriction is defined by precomposition:

$$(\widehat{\gamma}[f]_{\mathrm{raw}})(g) \coloneqq \widehat{\gamma}(f \circ g).$$

If $\mathcal{C}$ is definitionally associative and unital, then

$$\widehat{\gamma}[\mathsf{id}]_{\mathrm{raw}} \equiv \widehat{\gamma}, \qquad (\widehat{\gamma}[f]_{\mathrm{raw}})[g]_{\mathrm{raw}} \equiv \widehat{\gamma}[f \circ g]_{\mathrm{raw}}$$

hold definitionally.

Elements, Types, and Terms

A prefascist context Γ is decoded into the presheaf $x \mapsto \mathsf{El}\,\Gamma\,x$.

An element $\gamma : \mathsf{El}\,\Gamma\,x$ has two components:

- $\mathsf{run}\,\gamma$ is a raw thunk, assigning $\mathsf{run}\,\gamma\,f \in \Gamma_0(y)$ for every $f : y \to x$;
- $\mathsf{sat}\,\gamma\,f$ proves that every raw-thunk restriction is valid:

$$\mathsf{sat}\,\gamma\,f : \Gamma_1\big((\mathsf{run}\,\gamma)[f]_{\mathrm{raw}}\big).$$

Restriction in the decoded presheaf is written $\gamma[f]_{\mathsf{El}} : \mathsf{El}\,\Gamma\,y$, with

$$\mathsf{run}(\gamma[f]_{\mathsf{El}}) \equiv (\mathsf{run}\,\gamma)[f]_{\mathrm{raw}}.$$

A type $A \in \mathsf{Ty}(\Gamma)$ is the dependent version: $A_0(\gamma)$ gives raw data over γ, while A_1 validates dependent thunks. The type $\mathsf{El}\mathsf{T}\,A\,\gamma$ packages a dependent thunk together with proofs that all its restrictions are valid.

A term chooses $t(\gamma) \in A_0(\gamma)$ for every γ, compatibly with decoded restriction $\gamma[f]_{\mathsf{El}}$.

An Equivalence of Categories

Prefascist sets and their morphisms form a category $\mathsf{Pfs}(\mathcal{C})$.

The decoding above extends to an equivalence

$$\mathsf{Psh}(\mathcal{C}) \simeq \mathsf{Pfs}(\mathcal{C}),$$

assuming function extensionality.

Therefore, prefascist sets are an alternative presentation of presheaves with definitional restriction laws.

We want to build a CwF structure on $\mathsf{Pfs}(\mathcal{C})$.

Kaposi–Pujet

Our formalisation is inspired by Kaposi and Pujet's formalisation of the prefascist CwF_Π. This forms part of their broader strictification project. The recipe is available and well motivated in:

- Kaposi and Pujet, *Type Theory in Type Theory using a Strictified Syntax*, Sections 5.1–5.2;

Our Metatheory

Our metatheory can be viewed as a small fragment of Kaposi–Pujet's, inspired by observational type theory.

- Equality is homogeneous and Prop-valued: $x \equiv y : \mathrm{Prop}$ (proof irrelevant).
- We use three postulates: funext, coe, and coe-refl.
- Dependent transport and its computation rule have types

$$\mathrm{coe} : (P : A \to \mathrm{Set}) \to x \equiv y \to P(x) \to P(y),$$
$$\text{coe-refl} : \mathrm{coe}\, P\, \mathrm{refl}\, u \equiv u.$$

- coe-refl and the identity and associativity laws of the base category are registered as rewrite rules.
- We derive $\mathrm{coe}^{\mathsf{p}}$ and $\mathrm{funext}_{\mathsf{i}}$.

Main Difficulty I: Context Comprehension

The raw data of an extended context is a dependent pair:

$$(\Gamma \triangleright A)_0(x) \coloneqq \sum_{\gamma : \mathsf{El}\,\Gamma\,x} A_0(\gamma).$$

Its validity predicate is

$$(\Gamma \triangleright A)_1(\theta) \coloneqq \exists\,(e : \Gamma\text{-coh}(\theta)).\ A\text{-valid}(\theta, e).$$

- The coherence proof e shows that the first projections of θ assemble into $\gamma : \mathsf{El}\,\Gamma\,x$.
- The second projections are transported to form a dependent element over γ, which is required to be valid in A.
- This gives p, q, and pairing.
- Their equations compute definitionally except for context-comprehension η.

Main Difficulty II: Π-Types

An element of $\Pi_{a:A} B(a)$ is a dependent function sending each $a : A$ to an element of $B(a)$.

In our representation, an input $a : \mathsf{EIT}\, A\, \gamma$ is a record containing a dependent thunk and its validity proof. Quantifying directly over this record prevents the Π-substitution law from being definitional.

We therefore expose its two fields:

- $\mathsf{RunT}\, A\, \gamma$ describes the computational component;
- $\mathsf{SatT}\, A\, \gamma\, r$ states its validity;
- $\mathsf{packT}\, r\, s$ packages them back into $\mathsf{EIT}\, A\, \gamma$.

We define

$$(\mathsf{Pi}\, A\, B)_0\, \gamma \coloneqq (r : \mathsf{RunT}\, A\, \gamma) \to (s : \mathsf{SatT}\, A\, \gamma\, r) \to B_0(\gamma, \mathsf{packT}\, r\, s).$$

Mathematically, this is the same dependent function space; passing the two fields separately makes Π-substitution definitional.

Result

- We successfully formalise the prefascist CwF_Π over a base category whose categorical laws hold definitionally.
- All CwF_Π equations are definitional except for context-comprehension η, which is proved propositionally.
- A small OTT-inspired metatheory suffices for this construction.

Further Work

- Add more type formers, such as Booleans and identity types.
- Study which additional metatheoretic principles these type formers require.
- When the base category $\mathcal{C}$ is itself a CwF, extend the prefascist Yoneda embedding

$$\mathsf{Yo}_{\mathcal{C}} : \mathcal{C} \longrightarrow \mathsf{Pfs}(\mathcal{C})$$

to the corresponding weak CwF morphism.